WORKED SOLUTION

STEP III 2013, Question 5

Pure • Rationality and Coprimality

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q5

The key idea

Every step runs on one tool: if a number divides a product and is coprime to one factor, it divides the other factor. Put a rational number in lowest terms, clear denominators, and watch which side each factor has to land on. In part (ii) this traps the denominator b between two facts that can't both hold unless b is tiny.

We shall use one small fact repeatedly: *if p and q are coprime then so are p^s and q^t* for any positive integers s, t. Indeed, a prime dividing both p^s and q^t would divide p and q (a prime dividing a product divides one of the factors, which is the given result with a prime), contradicting coprimality.

Solution

Part (i)

From $q^nN = p^n$ we see that p^n divides q^nN. Since p and q are coprime, so are p^n and q^n, and the given result (with $a = p^n$, $b = q^n$, $c = N$) tells us that p^n divides N. So $N = kp^n$ for some positive integer k, as required.

Substituting back, $q^nkp^n = p^n$, so $q^nk = 1$. Both are positive integers, so $k = 1$ and $q^n = 1$, giving

$$q = 1 \quad (\text{and } N = p^n).$$

Now take any positive integers n and N and suppose $\sqrt[n]{N}$ is rational. It is positive, so we can write $\sqrt[n]{N} = p/q$ with p, q positive integers and the fraction in lowest terms, i.e. p and q coprime. Raising to the nth power, $q^nN = p^n$, and the work above gives $q = 1$. Hence $\sqrt[n]{N} = p$ is a positive integer. So $\sqrt[n]{N}$ is either a positive integer or irrational.

Part (ii)

We have $a^ad^b = b^ac^b$ with $\gcd(a, b) = \gcd(c, d) = 1$.

Showing $d^b = b^a$. The right-hand side shows that b^a divides $a^a d^b$. As a and b are coprime, b^a and a^a are coprime, so b^a divides d^b. Symmetrically, d^b divides $b^a c^b$, and d^b is coprime to c^b, so d^b divides b^a. Two positive integers that divide each other are equal:

$$d^b = b^a,$$

and then cancelling in the original equation gives $a^a = c^b$ as well.

If p is a prime factor of d, then p divides $d^b = b^a$, and a prime dividing a power of b divides b. So p is a prime factor of b.

Powers of p. Let p^m and p^n be the highest powers of p dividing d and b, with $m, n \geq 1$ by the previous paragraph. By unique factorisation, the highest power of p dividing d^b is p^{mb}, and that dividing b^a is p^{na}. Since $d^b = b^a$, these are equal, so $mb = na$ and

$$b = \frac{na}{m}.$$

? **Ask yourself** How does $b = na/m$ give a bound on p^n?

Rewrite it as $mb = na$: so b divides na. But b is coprime to a, so by the given result b divides n, and in particular $b \leq n$. Also p^n divides b, so $p^n \leq b$. Together,

$$p^n \leq b \leq n.$$

But $p \geq 2$ and $n > 0$, so the given fact $y^x > x$ (with $y = p$, $x = n$) says $p^n > n$. This contradiction shows that our assumption was impossible: d has *no* prime factor. Hence $d = 1$, so $b^a = d^b = 1$, and therefore

$$b = 1.$$

The final part

Let r be a positive rational with r^r rational. Write $r = a/b$ in lowest terms (a, b positive and coprime), and $r^r = c/d$ in lowest terms (c, d positive, since $r^r > 0$, and coprime). Then

$$\left(\frac{a}{b}\right)^{a/b} = \frac{c}{d} \quad \Longrightarrow \quad \left(\frac{a}{b}\right)^a = \left(\frac{c}{d}\right)^b \quad \Longrightarrow \quad a^a d^b = b^a c^b,$$

raising both sides to the power b and clearing denominators. This is exactly the situation of part (ii), so $b = 1$ and $r = a$ is a positive integer.

Checks

- Integer r really do work: $r = 2$ gives $r^r = 4$; and the conclusion $c^b = a^a$ with $b = 1$ just says $c = a^a$, as it should.
- A non-example: $r = \frac{1}{2}$ gives $r^r = 1/\sqrt{2}$, irrational, consistent with the result. In part (i), $\sqrt[3]{27} = 3$ and $\sqrt[3]{24}$ is irrational: both alternatives occur.
- A computer search over all coprime $a, b \leq 24$, $c < 200$, $d < 60$ found only solutions of $a^a d^b = b^a c^b$ with $b = d = 1$ (namely $a = 1, 2, 3$ with $c = a^a$).

Common slips

- Forgetting to say *why* p^n and q^n are coprime: the given result is about a and b coprime, so you need the powers to be coprime.
- Concluding $p^n \leq n$ from $b = na/m$ without the step $b \mid n$: it is coprimality of a and b that turns $mb = na$ into $b \mid n$.
- Writing $(a/b)^{a/b} = c/d$ but not insisting that c/d is in lowest terms; without $\gcd(c, d) = 1$, the step $d^b \mid b^a$ fails.

Beyond STEP

The given "coprime divisor" result is Euclid's lemma, usually proved from Bézout's identity, and it is the engine behind unique factorisation into primes. The first part is the general version of the irrationality of $\sqrt{2}$; the second is a first taste of transcendence theory, where the Gelfond–Schneider theorem shows, for instance, that $\sqrt{2}^{\sqrt{2}}$ is not even algebraic. Davenport's *The Higher Arithmetic* develops these foundations beautifully.

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